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Automated Classification and Size Estimation of Fetal Ventriculomegaly from MRI Images: A Comparative Study of Deep Learning Segmentation Approaches

K Gopikrishna*, N R Niranjan, Siddharth Maurya, V G Uday Krishnan, Simi Surendran

Department of Computer Science and Engineering, Amrita School of Computing, Amrita Vishwa Vidyapeetham, Amritapuri, India
amenu4cse20033@am.students.amrita.edu, amenu4cse20249@am.students.amrita.edu, amenu4cse20266@am.students.amrita.edu, amenu4cse20273@am.students.amrita.edu, simisurendran@am.amrita.edu

Abstract

Fetal ventriculomegaly is one of the major risks in prenatal diagnosis, which is an enlargement of the ventricles of the developing fetus's brain. Timely prediction of these brain disorders helps patients and healthcare providers make informed decisions and treatment plans. This paper investigates the classification of ventriculomegaly from MRI images using a deep learning approach. Segmentation of ventricles is performed using DeeplabV3+ and U-Net architectures and the accuracy, dice coefficient, and loss of these two approaches are compared. Additionally, a novel approach for size estimation was developed, utilizing contour detection to delineate ventricle boundaries and compute their areas by enclosing them within rectangles. A threshold-based classification method was then applied to differentiate between abnormal ventriculomegaly and normal cases. This research contributes to the assessment and comparison of the segmentation algorithms for aiding in the identification and classification of ventriculomegaly cases, offering potential implications for clinical diagnosis and treatment planning.

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Keywords: Ventriculomegaly; Deep Learning; U-Net; DeepLabV3+; Segmentation

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1. Introduction

Ventriculomegaly (VM), characterized by the abnormal dilation of the cerebral ventricles, poses significant challenges in prenatal diagnosis due to its varied etiology and potential neurological implications. Accurate detection and classification of ventriculomegaly[1] are crucial for timely medical intervention and appropriate management. Early prognosis enables healthcare professionals to plan treatments for neurological concerns and families to support the child.

This research addresses the critical need for a reliable diagnostic tool for ventriculomegaly, aiming to provide swift and accurate assessments amid challenges such as resource scarcity, diagnostic uncertainty, and time sensitivity. Leveraging the advantages of automated ventriculomegaly detection offers a promising solution to these challenges. It enhances monitoring and enables early diagnosis, thus mitigating the risks associated with this condition. Furthermore, it optimizes the utilization of scarce healthcare resources and services, making healthcare more accessible. By providing a reliable, precise, and consistent diagnostic approach, this automated system not only benefits healthcare professionals but also eases patient concerns and fosters a sense of assurance, ultimately contributing to improved patient outcomes and a more effective healthcare system.

Recent advancements in deep learning have facilitated the development of automatic classification of VM and non-VM cases from ultrasound or MRI imaging. segmentation algorithms capable of delineating ventricular structures from medical imaging data, offering potential avenues for automated diagnosis. This study focuses on the comparison of two prominent segmentation algorithms, DeeplabV3+ and U-Net, in the context of ventriculomegaly detection. By employing these algorithms, we aim to evaluate their performance based on standard segmentation metrics such as accuracy, Dice coefficient, and loss. Additionally, a novel approach for estimating ventricular size using contour detection and area computation is proposed, allowing for the classification of cases as ventriculomegaly (VM) or non-VM. The objective of this research is to provide a comprehensive assessment of these segmentation algorithms' efficacy in detecting ventriculomegaly and estimating the size of ventricular structures. The findings derived from this comparative analysis aim to contribute to the advancement of automated diagnostic tools for ventriculomegaly, potentially enhancing clinical decision-making processes and improving patient care in prenatal settings.

The remaining sections of this paper are organized as follows: Section 2 presents a review of works related to deep learning approaches for fetal brain abnormality prediction. In Section 3, we outline methods used for segmentation and algorithms for ventricle size estimation. Section 4 presents the segmentation and classification results followed by concluding remarks.

2. Related Work

This section reviews the works related to fetal brain analysis using deep learning approaches. Shao et al. [2] tackled the problem of accurately segmenting the ventricular system in patients with Normal Pressure Hydrocephalus (NPH) and proposed a new 3D U-Net method for ventricular segmentation in NPH patients, achieving high accuracy but with a small sample size. Vahedifard et al. [3] addressed the development of an AI model for automatic ventriculomegaly detection in fetal brain MRI, offering a promising approach to enhance diagnostic accuracy and clinical efficiency. However, the model's 3D reconstruction requirement and computational time pose limitations. Xie et al. [4] aimed to leverage deep-learning algorithms for classifying fetal brain ultrasound images as normal or abnormal, potentially reducing false-negative diagnoses. However, they faced data resource constraints and a relatively small sample size for abnormal cases.

Salehi et al. [5] focused on real-time fetal brain extraction from MRI scans using deep learning, demonstrating the superiority of their 2D U-net and autocontext approach over alternative methods. Milletari et al. [6] devised the Hough-CNN to automate MRI and ultrasound brain region segmentation, emphasizing its multi-modal, robust, and anatomically informed voting strategy. Challenges include smaller sample sizes and potential limitations due to 3D convolutions. Hesse et al. [7] pioneered a deep learning approach for subcortical segmentation in fetal ultrasound, employing a CNN to segment critical structures using few-shot learning. Challenges include sparse training labels and limited performance measure accuracy.

Liu et al. [8] automated the estimation of fetal lateral ventricular width from prenatal ultrasound images, enhancing measurement accuracy and consistency. However, lack of peer review and uncertainty in generalizability to other structures pose limitations. Liu et al. [9] addressed the limitations of traditional 2D deep learning methods for medical ultrasound analysis, emphasizing the unique challenges in ultrasound imaging. Obeid et al. [10] focused on predicting the need for temporizing cerebrospinal fluid drainage in newborns with intraventricular hemorrhage, highlighting the significance of ventricular shape evaluation on early ultrasound. Nabila et al. [11] addressed the need for automated measurement of Lateral Ventricle Ratio (LVR) in fetal ultrasound images to predict ventriculomegaly.

Many machine learning techniques have been applied in various real-time applications such as healthcare and wireless networks [12, 13, 14, 15, 16, 17, 18]. The proposed work was able to fill the research gap in the literature by addressing the need for a comprehensive comparison of deep learning-based segmentation algorithms and ventricle size estimation techniques for fetal brain abnormality prediction. While previous studies have focused on various aspects of fetal brain analysis using deep learning approaches, this paper specifically contributes to the assessment and comparison of segmentation algorithms for aiding in the identification and classification of ventriculomegaly cases. Additionally, the novel approach for ventricle size estimation using contour detection and area computation offers a new method for classifying cases as ventriculomegaly (VM) or non-VM, which was not extensively covered in the existing literature.

3. Deep Learning Approach for Ventricle Segmentation and Size Estimation

3.1. Dataset Description

To address the segmentation element of our work, we used a publically accessible MRI dataset obtained from Open Neuro[19]. This collection contains around 15 examples of segmented side ventricles, each with its mask. These masks were segmented by medical specialists from the MedSeg Team, a famous institution that provides segmentation services for CT or MRI datasets. This substantially alleviates any worries about the correctness of the masks used to train our model.

T1-weighted (T1W) scans are included in the collection, which are distinguished by their excellent anatomical detail. T1W scans show a clear distinction between different forms of brain tissue, including grey matter, white matter, and cerebrospinal fluid (CSF). This contrast is useful in identifying ventricular abnormalities because any enlargement or aberrant shape of the ventricles is more visible in T1W pictures. Furthermore, T1W scans are routinely used in clinical peerice, assuring their extensive availability in medical imaging facilities and simplifying our model's training under real-world situations. T1-weighted MRI increases the signal of fatty tissue while suppressing the signal of water, whereas T2-weighted MRI increases the signal of water. This distinction emphasizes the model's applicability and importance.

3.2. Data Preprocessing

The dataset generation process commenced with the acquisition of a publicly available MRI dataset from Open Neuro, comprising transventricular video files of MRI T1-weighted scans. These files, spanning approximately 15 cases, underwent meticulous segmentation by the MedSeg Team, focusing on detailed delineation of side ventricles along with their corresponding masks. To ensure seamless integration into various medical imaging pipelines, the transventricular video files were converted from .nii format to image format.

A critical phase in dataset curation involved a rigorous manual selection process, identifying high-quality images suitable for deep learning-based ventricle segmentation. From this careful selection, a final set of 753 images was curated, capturing a comprehensive spectrum of ventricular structures' nuances and variations.

In the process of replicating a ventriculomegaly case, a specific set of segmented masks was chosen to form ventriculomegaly cases. Using a manual approach, alterations were intricately drawn within the segmented masks to accurately reflect the distinctive features associated with ventriculomegaly. Utilizing the online software Pixlr, the ventricles within the masks were carefully enlarged, ensuring a realistic representation of ventricular expansion.

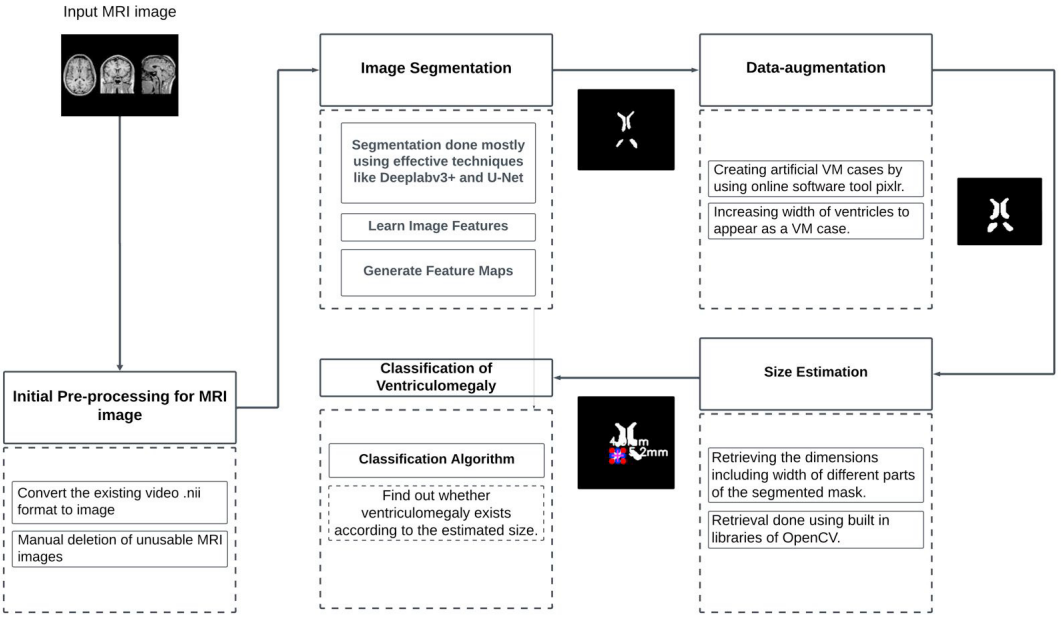


Fig. 1: Architecture of ventricle segmentation and VM classification

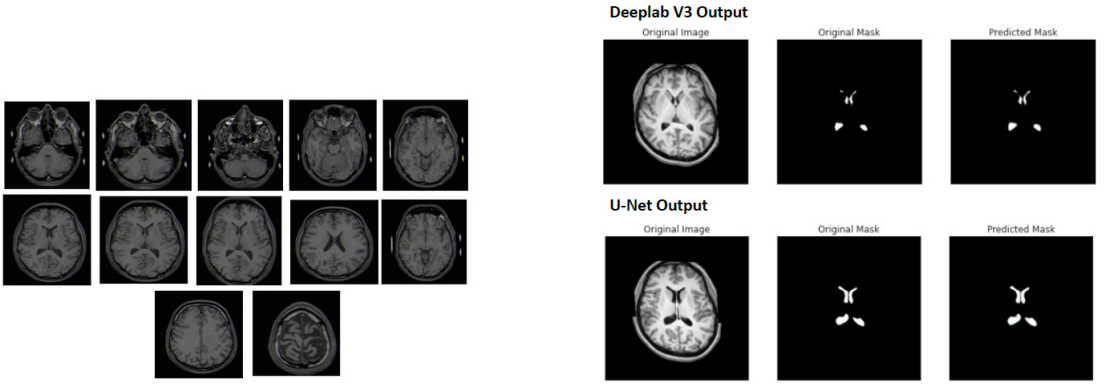
3.3. Segmentation and Size Estimation

The architecture of VM classification is presented in Figure 1. For the segmentation of fetal ventricles, we used Unet and DeepLabV3+. The U-Net architecture is a deep learning model that was specifically designed for medical image segmentation. It is a convolutional neural network (CNN) that consists of an encoder and a decoder, with skip connections between the two. The encoder extracts features from the input image, while the decoder reconstructs the segmentation mask from the extracted features. The U-Net architecture is well-suited for ventricle segmentation because it has several advantages over other CNN architectures: It can learn high-level features from the input image, while also preserving spatial information. This is important for ventricle segmentation, as the ventricles are small and complex structures. It can segment images of different sizes without the need for resizing. This is because the U-Net architecture uses a fully convolutional network, which means that it does not contain any fully connected layers. It is able to learn from a relatively small number of training examples. This is important for medical image segmentation, as labeled datasets are often limited. Figure 2a shows the samples obtained from the dataset and Figure 2b depicts the ventricles segmented using U-net and deeplab v3+ algorithms.

3.4. Size Estimation Algorithm

The size estimation algorithm utilizes computer vision techniques, implemented in Python with the OpenCV library, to estimate the dimensions of objects within medical images. The algorithm begins by preprocessing the input image, converting it to grayscale, and applying Gaussian blurring for edge detection. The function takes the edge-detected image, where the edges are highlighted following preprocessing procedures such as Canny edge detection, and identifies continuous curves along the object boundaries. The contours are these continuous curves. The edges of objects are located using changes in intensity in this extraction method. The contours are identified in the following image processing phases, resulting in a clear demarcation of the object borders.

The algorithm calculates the minimum area bounding box for each contour, orders its vertices, and computes midpoints and Euclidean distances between specific pairs of these midpoints. Importantly, a calibration factor based on a known-size reference object within the image is introduced ,using it as a reference to calibrate the dimensions of other objects in the scene. The reference object should have well-defined boundaries and stand out from the background.



(a) (b)

Fig. 2: (a) Sample MRI images extracted from the dataset (b) Segmentation of ventricles using Deeplab and U-net

The size of the reference object is assumed to be known in the code, and its dimensions are used to calculate a calibration factor. The measured distances between other detected objects are then converted into real-world dimensions using this factor. This factor is used to convert pixel distances into real-world dimensions, providing a precise estimate of object sizes in millimeters. The method's effectiveness is dependent on the proper selection of a reference object and meticulous calibration,facilitating the estimation of object sizes. The algorithm is demonstrated in a peerical context, showcasing its effectiveness in estimating the size of ventricles in MRI images, with potential applications in medical imaging research and diagnosis. Algorithm 1 shows the steps in size estimation of ventricles.

Algorithm 1: Ventricle Size Estimation from Segmented MRI

Data: Segmented MRI Image
Result: Estimated Ventricle Size

Generate the segmented ventricles from Unet and DeeplabV3+
while *contours exist in the image* **do**
  Identify and locate the contours of the ventricles in the segmented image.
  Compute the minimum area rectangle that encloses the contour as a bounding box around the ventricles.
  Calculate the midpoints of the bounding rectangles and mark lines between these midpoints to get a visual representation of the ventricles and their bounding rectangles.
  Identify the reference object based on its distinctive features or location
  Define the appropriate pixel-per-metric value as

$$\text{length(mm)} = \frac{\text{Number of pixels}}{\text{Pixels per mm}}$$

  Calculate the dimensions of the ventricles from the pixel per metric ratio
  if *estimated size is greater than 10mm* **then**
    Add to the probability of VM
  end
  else
    Search for VM again in the next contour
  end
end

3.5. Classification Algorithm

A threshold-based classification of ventricles is applied to the results of the size estimation process. Utilizing a predefined calibration factor, the algorithm assesses the sizes of extracted ventricles in MRI images. In this context, a case is classified as ventriculomegaly if the estimated size surpasses a specified threshold, typically set at 10mm. The algorithm employs Euclidean distances between midpoints of bounding boxes and subsequently scales these distances to real-world measurements. Algorithm 2 explains the steps in the threshold-based classification of VM cases. We used

Algorithm 2: Threshold-based VM Classification on the Segmented Ventricles

Data: Estimated Size in mm
Result: VM Prediction

```
while contours exist in the image do
    Compute the size of ventricles using Algorithm 1
    if estimated width greater than 10mm then
        result.push[1];
    end
    else
        result.push[0];
    end
end
if sum(result) == 0 then
    print("No Ventriculomegaly Detected");
end
else
    print("Chance of Ventriculomegaly");
end
```

Unet and Deeplab V3+ for segmentation of ventricles from MRI images. The factors taken into account for the comparison of the two algorithms are accuracy, dice coefficient, and loss. The accuracy, in the context of medical image segmentation, measures the proportion of correctly classified pixels or regions related to ventriculomegaly compared to the total number of pixels or regions. It is a fundamental evaluation metric used to gauge the overall performance of an algorithm. Loss functions quantify the disparity between predicted and ground truth segmentations during the training of neural networks. They guide the model towards minimizing errors and improving convergence. A lower loss value suggests that the model is effectively learning and aligning with the ground truth annotations.

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)\right]$$

The Dice coefficient is a metric used to measure the similarity between two samples, commonly employed in medical imaging tasks like ventriculomegaly detection. It calculates the overlap between the predicted and ground truth segmentations, providing insight into the spatial agreement between the identified regions and the actual affected areas. Higher values (closer to 1) indicate better segmentation accuracy.

$$dc = \frac{2 \times |A \cap B|}{|A| + |B|}$$

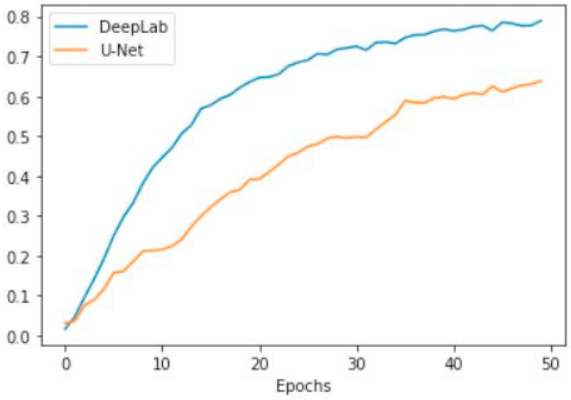


Fig. 3: Comparison of dice coefficient of segmentation by Unet and deeplab v3: Deeplab has a better dice coefficient and segmentation quality compared to U-Net

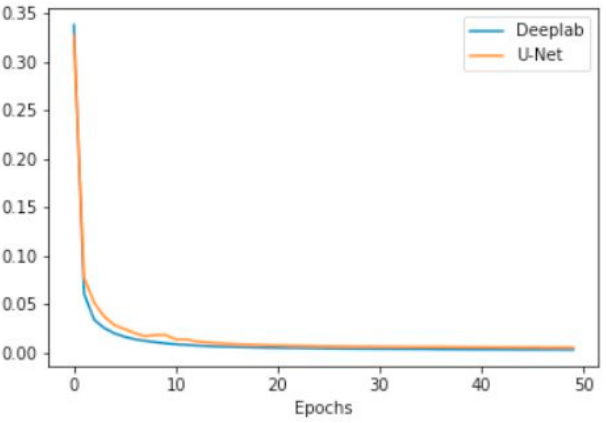


Fig. 4: Comparison of Loss Vs Number of epochs: Deeplab loss converges to a lower value faster than U-Net

Upon training the U-Net and DeepLab models with a batch size of 8 for 50 epochs, we obtained the performance patterns. The Dice coefficient, a metric crucial for evaluating segmentation quality, demonstrated significant disparity between the models. U-Net attained a Dice coefficient of approximately 0.6, indicating reasonable spatial agreement between predicted and ground truth segmentations. In stark contrast, DeepLab surpassed expectations, achieving a Dice coefficient exceeding 0.8 within the same training duration. This substantial difference in segmentation accuracy suggests DeepLab's superior capability in precisely delineating ventriculomegaly regions compared to U-Net. To obtain the Dice coefficient and visualize its progression, the model was configured to calculate this metric during training. This was achieved by compiling the model with the Dice coefficient as one of the specified metrics.To plot the Dice coefficient, the recorded metric values from the training process, specifically the Dice coefficient values per epoch, were extracted from the results object. Subsequently, Matplotlib, a Python plotting library, was employed to create a line plot. Figure 3 shows the variation of the dice coefficient with the number of epochs.

Moreover, the convergence of the loss function for both models was noteworthy. As training progressed, both U-Net and DeepLab showcased impressive convergence, with their respective loss values approaching close to 0.00 after 50 epochs. This convergence signifies effective learning and alignment with ground truth annotations, indicating successful training for both architectures. Figure 4 shows the comparison of loss that occurred in Unet and deeplab with the number of epochs.

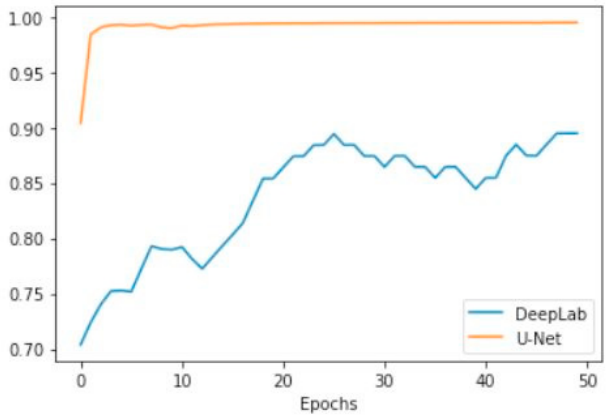


Fig. 5: Accuracy Vs Number of epochs of U-Net and Deeplab V3+ segmentation



Fig. 6: Sample output of size estimation from segmented ventricles

However, when evaluating the models based on accuracy metrics, a distinct discrepancy surfaced. U-Net exhibited exceptional accuracy, crossing the 0.98 mark, portraying high precision in classifying ventriculomegaly regions. On the other hand, DeepLabV3+ achieved an accuracy of approximately 0.89, demonstrating robust performance but slightly lower accuracy compared to U-Net. This discrepancy in accuracy metrics hints at a trade-off between segmentation quality, as indicated by the Dice coefficient favoring DeepLabV3+, and precise classification, where U-Net excelled. Figure 5 compares the accuracy obtained in both architectures.

Figure 6 shows a sample output from the size estimation algorithm. The entire width of the ventricle is clearly bounded by a minimum area rectangle and the width in millimeters is calculated by assuming a calibration factor of 2.6. From this estimated size the output is fed to the classification algorithm which then classifies the case to a VM case or not. Figure 7 shows the results from the threshold-based classification algorithm of VM. This algorithm obtains the size of the ventricles and checks if each white contour has a width over 10mm. If so the algorithm marks it as a highly probable VM, otherwise, it checks again in the next white contour.

4. Conclusion

This paper addressed the necessity for a dependable diagnostic tool in identifying ventriculomegaly in fetal brain ultrasound images within the domain of pediatric healthcare. Driven by challenges in this field, particularly the critical

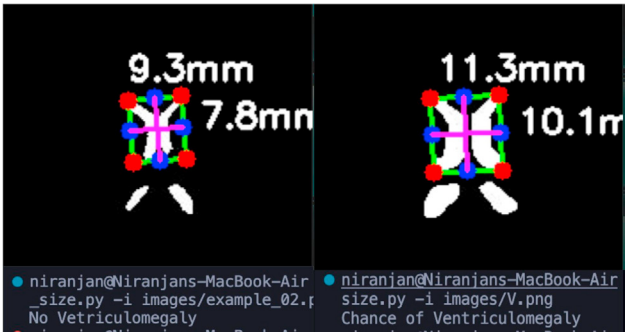


Fig. 7: A sample classification of VM and non-VM cases

need to promptly and accurately identify ventriculomegaly, our research rigorously examined two deep learning models U-Net and DeepLab V3+ for segmentation of fetal brain ventricles. DeepLab V3+ demonstrated notable precision in outlining ventriculomegaly regions, achieving a dice coefficient exceeding 0.8. Conversely, while U-Net showed a reasonable agreement in spatial terms with a Dice coefficient of approximately 0.6, its strength lay in accurate classification, attaining an accuracy score surpassing 0.98. The convergence of both models to close to zero loss after 50 epochs highlighted their effectiveness in learning during the training phase. Furthermore, we used a computer vision-based ventricle size estimation approach to classify VM and non-VM cases. This research contributes to the ongoing efforts to refine prenatal care and ventriculomegaly diagnosis through deep-learning approaches. The possible implementation of these models promises to improve diagnostic accuracy and thus advance the healthcare peerices in pediatric neurology.

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